

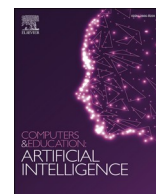


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Face value: How avatar identity shapes epistemic trust in AI-mediated learning

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ABSTRACT

AI tutors are becoming the new faces of digital classrooms, but these faces may determine who is trusted and who is ignored. Anthropomorphic avatars designed to increase engagement, may also activate social heuristics that reproduce offline patterns of bias. Yet little empirical research has examined how these identity cues shape epistemic trust and enacted reliance in AI-mediated learning. This study addresses this gap by systematically isolating their effects on learners' evaluative judgments and behavioral uptake in diverse learning contexts. Two experiments examined how avatar race, gender, and age shape trust in AI-mediated education. Study 1 (N = 102) used a within-subjects laboratory design with tightly controlled avatars; Study 2 (N = 294) adopted a between-subjects online design across varied instructional domains. Four outcomes, credibility, warmth, competence, and willingness to act, were measured alongside a behavioral index of epistemic uptake, defined as incorporating AI guidance into learners' own work. Across both studies, social identity cues strongly influenced evaluations and behavior. White avatars, and in STEM contexts Asian male avatars, were rated more credible and competent. Older Black female avatars faced compounded penalties across all measures. These perceptual hierarchies shaped behavior: participants were more likely to adopt guidance from avatars aligned with their racial ingroup or stereotypical expectations of expertise. Domain moderated these effects, with STEM and procedural tasks amplifying bias, while reflective and interpersonal tasks attenuated it. Our findings show that trust in AI tutors is as much about the messenger as the message. Avatar identity guides not only learners' perceptions but also their actions, underscoring the need for equity-aware avatar design and trust calibration mechanisms to ensure AI instruction empowers all learners. Building on these insights, we outline practical design recommendations to ensure that AI instruction amplifies learning rather than entrenches bias.

1. Introduction

The transmission of knowledge is the central nervous system of human culture. Unlike other species that rely primarily on genetic inheritance or individual trial-and-error learning, humans are obligate social learners. We depend on others to identify dangers, construct tools, and, in the modern era, interpret complex data and navigate digital environments. This reliance creates a fundamental vulnerability: the risk of misinformation, deception, or incompetence. To mitigate this risk while preserving the benefits of social learning, the human mind has evolved a filtering mechanism known as epistemic trust (ET), defined as "trust in the authenticity and personal relevance of interpersonally transmitted knowledge" (Fonagy & Campbell, 2017). Empirical work

suggests that ET is not merely a cognitive assessment of accuracy, but a deeply interpersonal stance shaping openness to socially communicated information (Aseeri & Interrante, 2021). In educational contexts, when ET is active, learners internalize and generalize information; when compromised, they may default to rigid mistrust or indiscriminate credulity (Desatnik et al., 2025).

As education increasingly unfolds within digital ecosystems, knowledge transmission is frequently mediated by AI-driven instructional agents, including avatars, embodied tutors, and conversational systems (Dai et al., 2023; Yan, Sha, et al., 2024). These agents function as epistemic interlocutors, shaping how learners engage with content, internalize information, and decide whether to act on guidance (Darvishi et al., 2024). Crucially, learners evaluate such systems not

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only in terms of epistemic accuracy, but through social cognition. Avatars are treated as quasi-social entities, and their outputs are often interpreted using heuristics evolved for human–human interaction (Jakesch et al., 2019; Nass & Moon, 2000). What might otherwise be peripheral social information is thus elevated into a central mechanism for evaluating knowledge claims. Research in social psychology shows that identity cues such as race, gender, and age systematically shape judgments of competence, warmth, and credibility (Fiske et al., 2002; Miller, 2019), thereby controlling whose knowledge is trusted and acted upon. In AI contexts, these biases are not neutral; they can reproduce entrenched asymmetries within educational systems (Baker & Hawn, 2022).

The socio-affective presence of AI avatars can influence learner engagement, motivation, and emotional orientation, implicitly shaping instructional outcomes. Yet trust in AI-mediated learning is rarely grounded in direct access to epistemic quality. Instead, it mostly emerges under conditions of limited transparency, where social and identity cues may substitute for verifiable reliability. In such environments, trust becomes a problem of inference under *epistemic opacity*. Although prior research has examined trust in educational AI and the social effects of avatars separately, these literatures remain conceptually disjoint in accounting for how avatar identity affects epistemic judgment in learning contexts. The present study addresses this gap by examining how identity-based perceptions of AI agents essentially shape epistemic trust and translate into differential reliance on AI-provided guidance. Epistemic trust is thus conceptualized not merely as a dispositional or attitudinal construct, but as a processual mechanism linking evaluation (competence, credibility, warmth) to intention (willingness to act) and behavior (epistemic uptake).

2. Literature review

2.1. Epistemic trust, trustworthiness, and trust calibration in AI-mediated learning

In this study, ET is conceptualized as learners' readiness to treat an AI-mediated instructional agent as a dependable source of knowledge and to rely on its guidance during learning activity (Afroogh et al., 2024; Heersmink et al., 2024; Sahebi & Formosa, 2025). Unlike general attitudes toward technology or interpersonal liking, ET concerns judgments of informational reliability, relevance, and usefulness under conditions where correctness cannot be independently verified and must instead be inferred from interactional cues (Legood et al., 2023; Shin, 2021; Yan, Sha, et al., 2024). As such, ET governs attention, acceptance, and enactment of guidance, linking evaluative perceptions to downstream behavioral responses (Nazaretsky et al., 2022; Viberg et al., 2025). Recent work on large language models (LLMs) and AI-based educational systems underscores the centrality of such trust, as learners must decide whether to accept, ignore, or act on AI-generated explanations and feedback, despite restricted access to the underlying models and data (Afroogh et al., 2024; Heersmink et al., 2024; Yan et al., 2025).

Rather than adopting dispositional measures such as the Epistemic Trust, Mistrust, and Credulity Questionnaire (ETMCQ), which capture stable orientations toward socially communicated knowledge (Campbell et al., 2021; Greiner et al., 2025; Milesi et al., 2024), the present study focuses on situated, source-specific evaluations of the instructional agent. In educational interactions, trust judgments are elicited by system properties, interface cues, and interactional dynamics, necessitating an operationalization that captures how learners evaluate a concrete source encountered in situ. This aligns with broader work in mediated trust, which has converged on interaction-level frameworks centring on source evaluation. Communication research conceptualizes credibility through expertise, trustworthiness, and goodwill (Flanagin & Metzger, 2007; McCroskey & Teven, 1999; Sundar, 2008, pp. 73–100); organizational models distinguish ability, benevolence, and integrity (Mayer et al., 1995); and human–automation research frames trust as an

acquired attitude, with calibration defined as alignment between human reliance and system reliability (Hoff & Bashir, 2015; Lee & See, 2004). To capture epistemic evaluation in instructional contexts, this study adopts the Muenster Epistemic Trustworthiness Inventory (METI), which conceptualizes trustworthiness along three dimensions: expertise, integrity, and benevolence (Hendriks et al., 2015). These map onto enduser judgments of pedagogical agents as competent, credible, and warm respectively, providing a framework suited to AI-mediated instruction. Importantly, such judgments are often shaped by social and identity cues (Davis et al., 2025; Nass & Moon, 2000; Shin, 2021; Yan, Sha, et al., 2024). However, while METI captures how trustworthiness is evaluated, it does not specify how these evaluations translate into learning behavior. In educational contexts, trust is meaningful only insofar as it translates into reliance through the incorporation of information into learners' reasoning or work, becoming miscalibrated when reliance is shaped by heuristics rather than epistemic quality (Afroogh et al., 2024; Heersmink et al., 2024; Lee & See, 2004; Okamura & Yamada, 2020; Yan, Sha, et al., 2024).

Building on this perspective, the present study examines alignment between stated willingness to act and observed epistemic uptake. Rather than measuring calibrated trust directly, willingness is incorporated as a self-reported indicator of intended reliance, while uptake is assessed as a behavioral outcome reflecting the integration of AI-provided guidance into task responses, enabling analysis of whether perceived trustworthiness translates into epistemic action and how such processes are shaped by avatar identity in AI-mediated learning.

2.2. Competence as perceived instructional authority

Perceived competence is a core indicator of ET in AI-mediated learning, capturing whether learners judge an instructional agent capable of performing the epistemic work that learning requires—producing guidance that is accurate, stable, and instructionally useable (Dorsch & Deroy, 2024; Alvarado, 2023; Cukurova, 2025). Within the METI framework, this dimension corresponds to trustee *expertise*, understood as the perceived capacity of a system to reliably execute domain-relevant tasks, even when supported by heuristic or pattern-based processes (Hendriks et al., 2015). Competence thus operationalizes expertise at the interactional level, determining whether an agent is experienced as sufficiently knowledgeable to justify epistemic reliance rather than as merely engaging (Heersmink et al., 2024; Omrani et al., 2022). Under this definition, competence precedes ET: if an agent is not judged capable of explaining concepts, diagnosing misconceptions, scaffolding problem solving, or evaluating responses, no defensible basis for epistemic uptake exists (Heersmink et al., 2024; Lucas et al., 2024; Ferrario, 2024).

This view locates trust in expectations of performance under uncertainty, where accuracy, robustness, consistency, and error behavior determine whether reliance is warranted (Afroogh et al., 2024; Durán & Pozzi, 2025; Ferrario, 2024), aligning with work that frames AI as an epistemic technology whose educational value derives from supporting inquiry and belief formation through performance rather than interactivity (Alvarado, 2023). In educational contexts, this judgment is especially salient because learners must determine whether an agent can teach, not merely produce plausible outputs (Bearman & Ajjawi, 2023; Yan, Sha, et al., 2024). Accordingly, competence addresses a distinct epistemic question from credibility: not whether information appears truth-connected, but whether the agent possesses sufficient domain expertise and pedagogical capacity to justify reliance. Empirical work shows that perceived competence predicts willingness to defer to feedback, persistence, and incorporation of AI guidance, particularly when performance is legible relative to expert benchmarks and known limitations (Labadze et al., 2023; Nazaretsky et al., 2022; Theunissen & Browning, 2022; Viberg et al., 2025). Related findings indicate that trust depends more on calibrated understanding of system capability than on general familiarity, with perceived capability functioning as a distinct

driver of educational use alongside trust (Chee, Ahn, & Lee, 2025; Lucas et al., 2024; Zhang et al., 2025; Dahri et al., 2024).

Under conditions of epistemic opacity, competence is inferred through observable proxies—such as response fluency, cross-interaction coherence, and behavioral consistency—rather than access to underlying reasoning processes (Bearman & Ajjawi, 2023; Theunissen & Browning, 2022). Error patterns and hallucinations function as diagnostic signals of epistemic performance, underscoring that competence is conditional and task-dependent rather than a fixed system property (Ji et al., 2023; Chelli et al., 2024). Over-reliance emerges when fluency is mistaken for competence, with consequences for decision quality and critical thinking (Yan et al., 2025; Zhai et al., 2024), rendering competence inseparable from epistemic vigilance. These evaluations are further shaped by socially interpretable cues: anthropomorphic realism, demographic congruence, and status signalling act as proxies for epistemic authority, influencing perceived expertise even when informational content is held constant (Moss-Racusin et al., 2012; Nowak & Rauh, 2008; Richter et al., 2025; Zhang et al., 2024). Avatars aligned with stereotypical authority are more readily judged as competent, whereas non-normative or marginalized representations may incur systematic penalties despite equivalent performance (Edwards et al., 2019; Ratan & Sah, 2015), in epistemically demanding domains (Etemad-Sajadi, 2016; Oh et al., 2016). Avatar identity thus becomes an infrastructural component of competence perception, shaping whether outputs are treated as viable candidates for epistemic uptake.

Nevertheless, competence remains necessary but not sufficient for ET: agents may appear highly capable while producing misleading or unstable outputs that learners struggle to detect, while high perceived capability can encourage uncritical acceptance (Chelli et al., 2024; Zhai et al., 2024; Zhang et al., 2025). This distinction (between performing epistemic work and warranting epistemic acceptance) motivates the transition to credibility as the next evaluative filter (Afroogh et al., 2024; Durán & Pozzi, 2025; Heersmink et al., 2024).

2.3. Credibility as epistemic reliability

Where competence enables reliance, credibility regulates trust. It constitutes the epistemic judgment through which learners assess whether an instructional agent's outputs are sufficiently accurate, truthful, and dependable to qualify as useable knowledge within a learning context (Durán & Pozzi, 2025). Across the trust-in-AI literature, credibility is grounded in beliefs about reliability and epistemic soundness, serving as the evaluative criterion through which reliance is warranted (Afroogh et al., 2024; Ferrario, 2024; Lukyanenko et al., 2022). As contemporary AI systems become increasingly agentic and self-directed, gains in performance are frequently accompanied by diminished opportunities for inspection and verification, rendering credibility not merely one antecedent among others, but the epistemic hinge through which trust becomes action-guiding.

Within educational AI, this logic is reflected in how trust is operationalized in practice. Empirical research shows that perceptions of reliability, correctness, and pedagogical value systematically shape learners' evaluations of AI-generated guidance (Nazaretsky et al., 2022; Viberg et al., 2025), while adjacent work frames trust in terms of perceived dependability, fairness, and output accuracy, positioning credibility-like judgments as the mechanism through which trust acquires practical force (Dahri et al., 2024; Rana et al., 2024). Within the present framework, credibility—aligned with METI's *integrity*—thus represents the learner's evaluative verdict on whether an instructional agent constitutes a legitimate epistemic source, grounded not in presumed intent nor mere computational performance, but in perceived epistemic reliability and responsible knowledge deployment (Hendriks et al., 2015; Durán & Pozzi, 2025; Ferrario, 2024). Credibility judgments become more contingent under conditions of limited epistemic transparency, where epistemic quality cannot be directly assessed. Learners cannot inspect model reasoning or audit training data, and must instead

rely on indirect cues to infer reliability. This asymmetry of epistemic access has been identified as a key driver of over-reliance on AI, where outputs may be treated as satisficing, with downstream consequences for decision quality, critical thinking, and epistemic vigilance (Heersmink et al., 2024; Zhai et al., 2024). Experimental evidence shows that internally coherent or expert-like explanations can substantially elevate trust even when users cannot determine whether they are being guided toward truth (Pareek et al., 2024), underscoring that perceived credibility is shaped as much by interpretive scaffolding as by verifiable correctness.

In instructional contexts, these judgments are further structured by how agents signal epistemic integrity and authority through their social and communicative presentation. Professionalized appearance, institutional framing, and expert-aligned interaction styles function as cues of truthfulness and responsible knowledge production, influencing whether outputs are interpreted as reliable or merely plausible (Baake et al., 2025; Wu et al., 2024; Xu et al., 2025). Verbal immediacy and supportive phrasing can elevate perceived trustworthiness independently of epistemic quality (Abendschein et al., 2024; Brummernhenrich et al., 2025), while identity and role cues shape whether agents are treated as dependable epistemic sources or as provisional communicators (Davila, 2025; Edwards et al., 2019). Credibility is therefore central to AI-mediated learning precisely because it governs whether guidance is treated as knowledge, while remaining systematically exposed to bias through presentation and identity cues (Afroogh et al., 2024; Viberg et al., 2025). All things considered, even competent and/or epistemically credible systems do not guarantee engagement or appropriate reliance; learners must first be willing to enter (and sustain) the interaction through which guidance is encountered.

2.4. Warmth as Social-Affective gating of epistemic trust

At this point, epistemic trust cannot be understood solely in terms of technical capability and informational reliability. Before competence can be exercised and credibility evaluated, learners must first experience an instructional agent as socially approachable, supportive, and safe to engage with. Warmth captures these perceptions of social intent—approachability, supportiveness, and interpersonal responsiveness—and reflects whether an agent is experienced as benevolent rather than indifferent or adversarial (Fiske et al., 2002; Richter et al., 2025). Within the Stereotype Content Model, warmth constitutes a primary dimension of social evaluation because it addresses the first-order question of intent (“friend or foe?”), shaping receptivity prior to competence assessment (Fiske, 2018; Fiske et al., 2002). Parallel trust models similarly position benevolence as a foundational component of trustworthiness and a prerequisite for willingness to accept vulnerability (Mayer et al., 1995). Together, these frameworks converge on a shared logic: learners evaluate not only whether an agent can produce adequate outputs, but whether it fosters perceptions of psychological safety, and this judgment conditions epistemic uptake.

This gating function is particularly salient in learning contexts, where engagement with guidance often entails exposure of uncertainty and error (Bearman & Ajjawi, 2023; Kuhail, Alturki, Alramlawi, & Alhejori, 2023; Zhai et al., 2024). Under such conditions, warmth operates as a receptivity mechanism, shaping whether instructional input is attended to and enacted prior to any evaluation of correctness (Qin et al., 2020; Richter et al., 2025; Seitz, 2024). Experimental work manipulating warm versus competent agents shows that warmth enhances affective connection and engagement, whereas competence primarily supports task execution, indicating distinct but complementary pathways (Richter et al., 2025). Related studies of embodied tutors similarly show that social expressiveness increases engagement even when learning performance remains unchanged (Ackermann et al., 2025; Troshani et al., 2021), while research on artificial empathy demonstrates that such responses elevate perceived warmth and intensify willingness to rely on conversational agents in high-stakes contexts

(Ma et al., 2025; Seitz, 2024).

Warmth judgments are systematically shaped by avatar identity cues and interactional design features that signal benevolence, empathy, and relational availability. Communication style, affective phrasing, facial expressiveness, voice characteristics, and demographic presentation function as heuristics through which learners infer whether an agent is supportive and responsive (Fiske et al., 2002; Khadpe et al., 2020; Richter et al., 2025; Zhang et al., 2024). Avatars perceived as socially familiar or emotionally expressive tend to elicit stronger affective connection, whereas socially distant or ambiguously framed representations are more likely to be experienced as indifferent, even when instructional content is equivalent (Ackermann et al., 2025; Richter et al., 2025). These effects are further structured by gendered and cultural stereotypes that associate certain identities with nurturance and care, and others with authority or emotional distance, producing differentiated trust profiles across learner–avatar configurations (Fiske et al., 2002; Kervyn et al., 2015). Cross-domain evidence reinforces this pattern: warmth impressions formed during single interactions can exert durable effects on institutional trust and willingness to cooperate, while adoption in relationally demanding contexts remains contingent on credible signals of benevolent intent (Belanche et al., 2021; Hansen, 2022; Kolbl et al., 2020; Peng et al., 2022). Within educational contexts, trust is similarly shaped not only by technical functionality but by perceived helpfulness, interaction quality, and contextual sensitivity (Qin et al., 2020; Kuhail et al., 2023). Warmth thus regulates affective openness to instructional guidance, shaping persistence and responsiveness to feedback, and modulating the influence of competence and credibility. Still, these evaluations become educationally consequential only when they translate into reliance, positioning willingness to act as a logical next stage in the process.

2.5. Willingness to act and epistemic uptake: from perception to behavior

Across human–automation interaction, human–AI collaboration, as well as broader educational technology research, trust is increasingly conceptualized not as a static attitude but as a process linking evaluative judgments to patterns of reliance in situated activity. Contemporary models explicitly distinguish between trust as an internal psychological state, trust as intention to rely, and trust as enacted reliance behavior, emphasizing that favorable perceptions do not automatically translate into system use in decision-making or learning practice (Hoff & Bashir, 2015; Lee & See, 2004; Okamura & Yamada, 2020). In AI-mediated learning, this distinction is significant. Learners may judge an instructional agent as competent, credible, and socially supportive, yet still ignore, selectively reinterpret, or override its guidance when engaging with concrete tasks. Within this process view, willingness to act represents the transition from epistemic evaluation to intentional readiness. It captures learners' self-reported inclination to use AI-generated explanations, recommendations, or feedback as *knowledge-relevant guidance*; that is, whether they anticipate deferring to it in problem solving, writing, revision, etc., as opposed to treating it as background information (Hoff & Bashir, 2015; Shin, 2021). In human–AI interaction research, such intention measures are routinely treated as proximal antecedents of reliance, mediating the path from perceived trustworthiness to subsequent use (Zhang & Lian, 2025).

However, a robust finding across automation and decision-support research is that intention is not equivalent to reliance. Users may report high trust and strong willingness to follow system outputs while simultaneously engaging in inconsistent, minimal, or strategically selective use in practice (Dzindolet et al., 2003; Lee & See, 2004; Schemmer et al., 2023). Educational settings exhibit the same vulnerability: students can express confidence in and appreciation of AI tutors or feedback systems while continuing to rely primarily on prior beliefs, peers, or trial-and-error strategies when completing tasks (Nazaretsky et al., 2022; Tosell et al., 2024; Viberg et al., 2025). These gaps reflect the well-established limits of self-report measures for capturing situated

epistemic behavior under conditions of time pressure and/or cognitive load (Stadler et al., 2024).

For this reason, recent research increasingly treats uptake as a decisive indicator of epistemic reliance. In the epistemic cognition and science education literature, learning is understood as epistemically consequential only when externally provided information is appropriated into learners' own standards of reasoning, justification, and hands-on knowledge construction (Barzilai & Chinn, 2018, 2024; Chinn et al., 2011; Sandoval, 2014). Building on this work, we conceptualize epistemic uptake as the threshold at which instructional guidance ceases to be a mere external prompt and is instead reconstituted as an internal epistemic commitment. In applied learning contexts, such appropriation becomes visible through learners' epistemic actions, including revising answers in line with feedback, adopting recommended solution strategies, or integrating external explanations into written justification (Lin et al., 2023; Ouyang et al., 2023). Studies in AI-supported learning environments further indicate that instructional support becomes meaningful only when learners actively re-use and adapt it in their own inquiry processes and when it is reflected in epistemically accountable productions (Liu et al., 2022; Nieminen, Haataja, & Cobb, 2025). Epistemic uptake therefore provides a theoretically grounded indicator that complements perceptual and intentional measures of reliance; a definitive marker of whether instructional guidance has been truly internalized (Holmes et al., 2022; Küper & Krämer, 2025). But even epistemic uptake is not synonymous with compliance. Research on calibrated trust emphasizes that appropriate reliance involves selectively accepting, questioning, or rejecting system outputs in accordance with their objective reliability and contextual relevance (Afroogh et al., 2024; Lee & See, 2004; Okamura & Yamada, 2020). In educational contexts, indiscriminate adoption of AI-generated content can undermine critical thinking and epistemic agency, while blanket rejection can foreclose potential instructional benefits (Al-Zahrani, 2024). Thus, epistemic uptake can provide a window into how learners negotiate epistemic authority in practice; whether reliance reflects critical evaluation or is driven primarily by surface-level convenience.

These premises highlight the importance of examining epistemic uptake independently. Willingness to act captures declared readiness to accept an instructional agent as a provisional epistemic authority; epistemic uptake captures whether this readiness is enacted through the integration of AI-provided information into production. Together, they enable ET to be pragmatically scrutinized in terms of (mis)alignment. This is especially important in AI-mediated learning environments characterized by socially persuasive interfaces, where strong impressions of competence, credibility, and warmth may form without reliable access to underlying accuracy or evidential grounding, increasing the risk of both over-reliance and excessive skepticism toward valuable instructional support (Dzindolet et al., 2003; Yan et al., 2025; Zhai et al., 2024). By extending evaluative dimensions with intentional and behavioral indicators, this study responds directly to long-standing calls to ground trust assessment in observable reliance behavior rather than in self-report alone (Lee & See, 2004; Hoff & Bashir, 2015; Küper & Krämer, 2025). Specifically, willingness is positioned as an extension of METI beyond perceptual evaluation, whereas uptake captures the realization of reliance in task performance, enabling direct comparison between *intended* and *enacted* epistemic reliance.

To sum up, competence, credibility, warmth, and willingness to act form a nested hierarchy of trust—one that determines whether an AI agent is gated, evaluated, and ultimately accepted as a dependable source of knowledge. By mapping the trajectory from initial social perception to epistemic action, this framework clarifies whether AI-mediated sociality produces a façade of compliance or supports the agentic appropriation of knowledge, thereby establishing the conditions under which AI guidance becomes epistemically consequential for learning.

3. Research gaps and study overview

Synthesizing these trajectories reveals various empirical and conceptual gaps in the literature. This review foregrounds underexplored questions at the intersection of trust calibration, identity cue processing, and sustained pedagogical adoption, issues that seem to remain insufficiently addressed by existing models, despite the growing prominence of AI-mediated learning. First, most prior scholarship on avatar credibility fails to engage with instructional contexts, where trust must translate into behavioral intention, metacognitive engagement, and knowledge transfer (Shin, 2021). Second, identity attributes such as race, gender, and age are often examined in isolation, limiting insight into the intersectional configurations that shape real-world bias and social perception (Jussim, 2017). Third, although Social Identity Theory predicts stronger trust toward agents perceived as ingroup members (Banks & Bowman, 2016; Tajfel et al., 2001, pp. 94–109), experimental work on learner–avatar identity alignment remains limited in educational settings. Finally, willingness to act, a core behavioral indicator of pedagogical trust and instructional transference, remains significantly underexamined relative to more commonly measured constructs like likability or perceived credibility (Fink et al., 2024; Teng, 2010).

This paper addresses these gaps through two complementary, theory-driven experiments that examine how avatar identity attributes (race, gender, age) and participant demographics interact to shape learners' judgments across four outcome dimensions: credibility, warmth, competence, and willingness to act. Drawing on the Stereotype Content Model (Fiske et al., 2002), Social Identity Theory (Tajfel et al., 2001, pp. 94–109), and the Computers as Social Actors framework (Nass & Moon, 2000), we investigate how social biases are reproduced, attenuated, or refracted in AI-mediated instruction. Study 1 employs a within-subjects design ($N = 102$) with tightly controlled avatar stimuli to isolate the effect of identity cues under conditions of high experimental control. Study 2 adopts a between-subjects, ecologically valid design ($N = 294$) using a demographically diverse online sample and domain-varied content (e.g., STEM vs. interpersonal) to assess generalizability and context sensitivity. Both studies leverage validated psychometric instruments, stratified sampling, and content-equivalent instructional scripts. Together, they provide a high-resolution, empirically grounded account of identity-based evaluations in digital learning environments.

Based on the aforementioned problem statement, we pose the following research questions:

RQ1. How do avatar race, gender, and age influence learners' perceptions of credibility, warmth, competence, and willingness to act on guidance?

RQ2. To what extent does identity alignment between learner and avatar (e.g., racial or gender ingroup/outgroup match) modulate these evaluations?

RQ3. Are these effects domain-dependent, and do intersectional avatars experience any additive or multiplicative credibility penalties?

By illuminating how entrenched social heuristics persist, shift, or are disrupted in avatar-mediated instruction, this research contributes to the design of equity-aware learning systems, the development of inclusive pedagogical agents, and to ongoing theoretical debates on ET in human–AI educational interaction. Methodologically, the study extends existing approaches to trust in AI-mediated learning by incorporating a behavioral indicator of epistemic uptake, allowing examination of how trust judgments relate to enacted reliance beyond self-report.

4. Methodology

4.1. Research design

Numerical results provide objective evidence related to the outcomes of reported research. However, such experimental outcomes always involve some randomness and so produce ranges of values. Statistical analyses are the only way to distinguish such randomness from meaningful responses that can support the study's conclusions. To investigate how surface-level social identity cues embedded in pedagogical avatars influence learners' trust-related judgments and instructional behaviors, we conducted two complementary experiments integrating high experimental control with ecologically grounded task design. Both experiments operationalized avatar race, gender, and age as independent variables, while perceptual and behavioral outcomes were measured across four epistemically relevant constructs: credibility, warmth, competence, and willingness to act. The research design was constructed to address three interrelated questions concerning (1) the main effects of avatar identity on evaluative and behavioral responses (RQ1), (2) the moderating influence of participant–avatar identity alignment (RQ2), and (3) domain-sensitivity and intersectional effects (RQ3). We focused on human-like avatars (rather than non-anthropomorphic agents) because the study aimed to examine how social identity cues shape ET processes, which presupposes the presence of visibly human demographic markers. This design choice follows established work showing that social categorization, stereotype activation, and ingroup–outgroup dynamics are most reliably elicited when agents are perceived as socially human-like rather than abstract or purely functional (Glikson & Woolley, 2020; Seaborn, 2025).

Study 1 employed a within-subjects counterbalanced design ($N = 102$), wherein each participant interacted with eight avatars whose identity attributes varied systematically across trials. This structure allowed for intra-individual comparisons under tightly controlled conditions and minimized between-subject variability, thereby enhancing statistical power and construct discrimination. Study 2 implemented a between-subjects randomized design ($N = 294$), wherein participants were exposed to a single avatar randomly assigned from the same identity set. This configuration maximized ecological validity and permitted the analysis of domain-specific effects under more naturalistic conditions. Together, the studies offer a complementary evidence base, triangulating internal and external validity concerns as recommended in experimental education research (Cook et al., 1979; Shadish et al., 2002).

To ensure stimulus invariance across conditions, all avatars were constructed using the HeyGen Interactive Avatar platform, which supports dynamic generation through a text-to-video prompt architecture (see Fig. 1). This allowed precise control over multimodal output features, including synchronized lip movement, prosody, affective neutrality, and visual realism, while varying only identity-relevant parameters (e.g., skin tone, facial structure, apparent age, vocal register). Each avatar was rendered in a neutral position, with standardized attire, eye gaze, and speaking cadence. Verbal content was generated from a fixed instructional knowledge base and delivered verbatim across all identity conditions. The resulting agents were equivalent along semantic, paralinguistic, and gestural dimensions, enabling valid attribution of observed effects to the manipulated social identity variables, a core criterion in stimulus-control experiments involving social cognition (De Visser et al., 2016; Nowak & Fox, 2018).

The avatar interaction occurred during an academic assessment task presented as a routine digital learning module. Participants completed a brief domain-diverse test and were offered optional access to a virtual assistant on each item. The assistant was framed as an integrated support feature of the assessment environment, not as the object of evaluation, thus reducing demand characteristics and promoting ecological authenticity in learner behavior (Aronson et al., 2010; Liao et al., 2016; Nass & Moon, 2000). Participants could invoke the avatar's response via

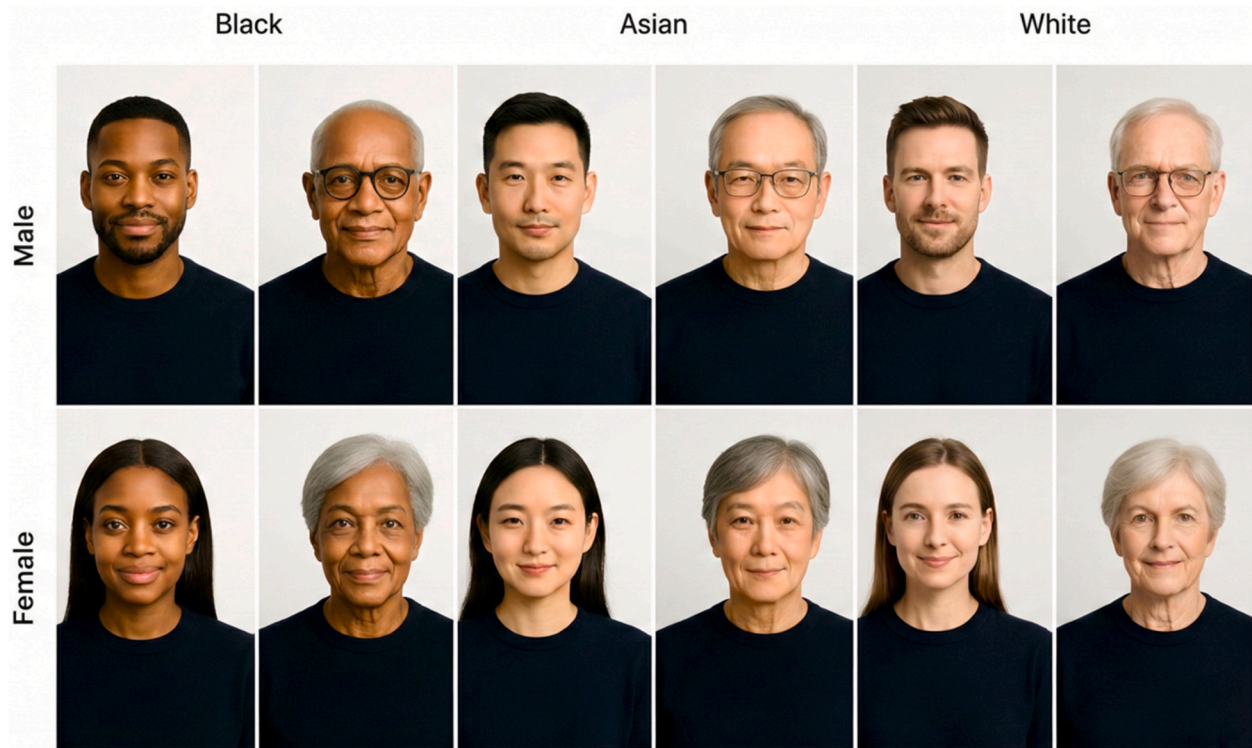


Fig. 1. Avatar Stimulus Set ($N = 12$); avatars varied systematically by race/ethnicity, gender, and age group, with all other perceptual features held constant. Identity categories reflect self-ascribed social constructs used in alignment with the experimental design and were not disclosed to participants.

a help prompt, receive a pre-scripted explanation, and then proceed to answer the item. No reference was made to avatar identity relevance, and participants were naïve to the study's hypotheses. In addition to conventional perceptual ratings, we operationalized behavioral trust through a novel outcome measure termed epistemic uptake. This construct captured the extent to which learners integrated avatar-delivered content into their own written responses, either through direct quotation or semantically equivalent paraphrasing. All test responses were logged and analyzed using automated string-matching and paraphrase detection algorithms, enabling high-resolution quantification of instructional influence at the point of action. By aligning this behavioral outcome with trust theory and pedagogical relevance, we provide a measure that is not only ecologically valid but also inferentially diagnostic of trust transfer, the point at which a belief in agent credibility is enacted as learner behavior.

4.2. Participants

Two independent samples participated in this study: one drawn from a controlled university-based setting (Study 1) and the other from a more ecologically varied online context (Study 2). Both samples were stratified to ensure demographic diversity across race/ethnicity, gender, and age, reflecting the identity dimensions manipulated in the avatar stimuli. Eligibility criteria included fluent English proficiency and active enrollment or recent participation in a higher education program, consistent with the study's educational framing.

Study 1 recruited 102 undergraduate and postgraduate students (Mean age = 23.7 years, $SD = 4.6$; 56.9% female) from a large, urban university. Participants were recruited via campus mailing lists and departmental bulletin boards. The sample was purposefully stratified by self-identified race (White, Black, Asian), gender (male, female), and age band (<25; 25+), with balanced representation across categories to support within-subject identity alignment analysis. All participants completed the study in person, under supervised conditions, in a controlled laboratory setting equipped with desktop workstations and

noise-isolated partitions. Each session lasted approximately 40 min. Participants took part voluntarily, did not receive compensation, and ethical approval was obtained from the university's institutional review board (IRB).

Study 2 recruited 294 participants (Mean age = 26.1 years, $SD = 5.9$; 53.4% female) from a vetted online research participant pool using stratified quota sampling to match Study 1's demographic targets. Recruitment was conducted via academic research platforms and professional participant services (e.g., Prolific, Qualtrics Panels), with screening filters applied to exclude bots, non-native English speakers, and individuals with duplicate IPs or prior exposure to avatar studies. All participants completed the experiment remotely, using their personal devices, via a secure online platform. Attention checks and engagement metrics (e.g., time-on-item, response variance) were embedded throughout the task to ensure data integrity (aka overcome any potential latency or inconsistency).

Participants in both studies provided informed consent, were blind to the study's aims, and received no compensation. Data collection and storage protocols adhered to institutional ethical standards, with full anonymization, secure encryption, and compliance with data protection regulations. No personally identifying information was collected. Ethics approval was granted by the university's institutional review board. The combined sample ($N = 396$) reflects a demographically diverse population representative of the target population for AI-mediated learning environments in higher education contexts (see Fig. 2).

4.3. Measures

To capture how learners appraised and responded to avatars varying systematically in race, gender, and age, we assessed four outcome constructs central to pedagogical trust: credibility, warmth, competence, and willingness to act. These constructs were selected based on their conceptual relevance to social cognition and human-agent interaction, as well as their recurring operational alignment with trust formation models across both instructional and affective dimensions (Glikson &

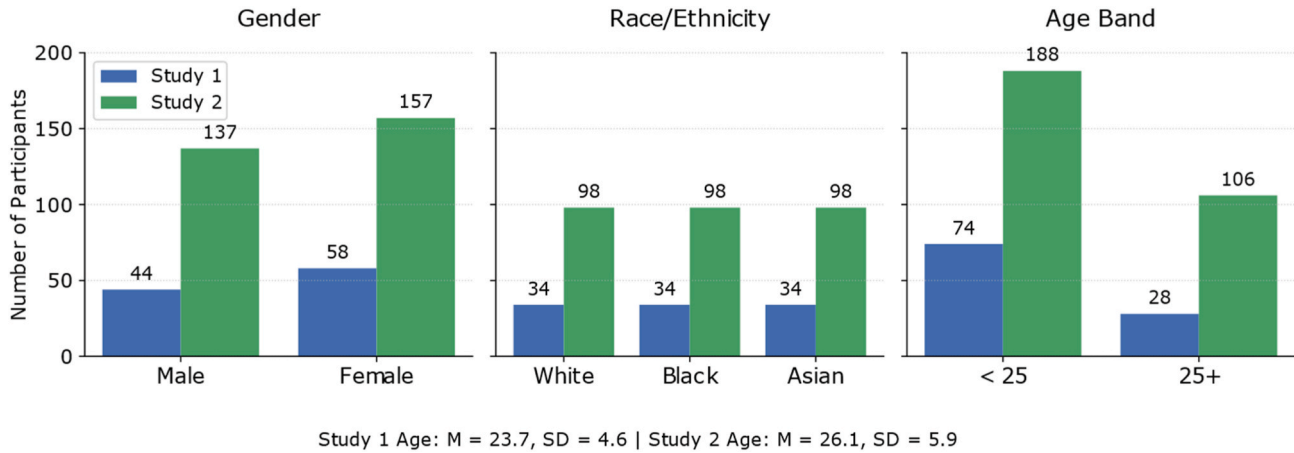


Fig. 2. Distribution of participants by self-identified race/ethnicity, gender, and age group. Labels such as “Black,” “White,” and “Asian” are used in accordance with conventions in educational and social cognition research, and refer to self-ascribed social identity categories rather than biological or essentialist classifications.

Woolley, 2020; Mayer et al., 1995; Shen et al., 2009). Together, they provide a multidimensional account of both evaluative and behavioral trust, extending beyond mere affective response to measure instructional transference.

All perceptual judgments were assessed through 5-point Likert-type items (1 = Strongly Disagree, 5 = Strongly Agree) administered immediately after each avatar interaction. This protocol preserved ecological realism while minimizing reflection-based distortion or demand (retrospective) bias. Initial item sets were subjected to exploratory factor analysis (EFA) using Promax oblique rotation to test discriminant validity across the four constructs. Items demonstrating factor cross-loadings greater than .40 or item-total correlations below .50 were excluded or reformulated. Final internal consistencies across constructs were satisfactory (Cronbach’s $\alpha = .84-.89$), and individual items showed sufficient ordinal dispersion for inferential modeling (see Table 1).

Credibility was operationalized as perceived epistemic reliability, the extent to which the avatar’s explanation was regarded as factually accurate, impartial, and cognitively dependable. Unlike interpersonal likability metrics, credibility here was conceptualized through the lens of source evaluation in digital and educational contexts, emphasizing perceived truthfulness and objectivity over relational affinity (Fink et al., 2024; Metzger & Flanagan, 2013). Items reflected judgments about the avatar’s capacity to offer trustworthy guidance within the domain of inquiry, decoupling surface presentation from inferential substance.

Warmth captured perceptions of the avatar’s social approachability, empathy, and benign intent. Rooted in the Stereotype Content Model (Fiske et al., 2002) warmth functions as a core affective dimension

shaping social heuristics across human and artificial agents alike. In AI-mediated learning, warmth is associated with increased rapport, parasocial engagement, and affective receptivity, especially in contexts where learners navigate uncertainty or feedback (Edwards et al., 2019; Nowak & Rauh, 2008). We included warmth to assess whether identity cues activated stereotypic expectations of interpersonal disposition, which may bias trust attribution independently of content quality.

Competence referred to perceived domain-relevant expertise and cognitive fluency. The construct reflects judgments of the avatar’s instructional authority, its ability to explain, clarify, and facilitate understanding within the task at hand. Drawing on work in intelligent tutoring systems and social robotics, competence has been shown to influence retention, epistemic deference, and user persistence (Baylor & Kim, 2005; Glikson & Woolley, 2020). Importantly, in our operationalization, competence was distinguished from general intelligence or likability; it focused on task-oriented evaluation of instructional adequacy.

Willingness to act, represented the behavioral manifestation of trust, i. e., whether learners intended to or actually did use the avatar’s input in their subsequent decisions. To capture this construct with both attitudinal and behavioral fidelity, we adopted a dual-channel measurement model. First, self-reported behavioral intention was measured using three Likert items that probed participants’ willingness to apply the avatar’s guidance in future learning situations. Wording was carefully designed to avoid hypothetical framing or acquiescence bias (e.g., “I would rely on this avatar’s explanation when solving similar problems”). Item responses were aggregated into a composite index ($\alpha = .85$). Second, we assessed behavioral uptake through an original measure of epistemic transference. Each written answer was algorithmically compared to the scripted explanation using Sentence-BERT embeddings (Reimers & Gurevych, 2019), with alignment quantified via cosine similarity and interpreted as high-probability content overlap. A threshold of .75, consistent with semantic textual similarity benchmarks and typical precision–recall tradeoffs in paraphrase detection, was applied (Cer et al., 2017); sensitivity checks across adjacent thresholds (.70–.80) yielded consistent classification patterns. Flagged responses were subsequently reviewed by two trained human coders using a structured rubric; inter-rater agreement was high ($\kappa = .91$). To facilitate ordinal analysis, uptake scores were trichotomized into:

0 – No Uptake: No conceptual or lexical alignment with the avatar’s guidance; response reflects independent reasoning or unrelated content.

1 – Partial Uptake: Surface-level alignment through paraphrased fragments or loosely related concepts; semantic overlap present but incomplete.

2 – Full Uptake: Clear transference of the avatar’s explanation via near-verbatim reproduction or semantically equivalent paraphrase.

Table 1
Summary of construct validity and reliability for trust measures.

Construct	Definition	Items	α	EFA Notes
Credibility	Perceived reliability, accuracy, and truthfulness of the avatar	4	.89	Loadings: .58–.72; no cross-loadings > .40; item–total $r \geq .52$; variance explained = 58%
Warmth	Perceived friendliness and interpersonal receptiveness	3	.87	One item excluded (cross-loading = .42); retained loadings $\geq .59$; variance explained = 54%
Competence	Perceived knowledge and pedagogical capability	4	.84	Loadings: .61–.73; no cross-loadings > .35; communalities $\geq .50$; variance explained = 59%
WTA	Self-reported intention of acting on avatar suggestions	3	.86	Loadings $\geq .60$; item–total $r = .51-.57$; variance explained = 56%

This hybrid method enabled a fine-grained distinction between superficial alignment and deep instructional influence, thereby offering a behaviorally grounded index of whether ET resulted in observable knowledge application. Collectively, this multi-method measurement strategy supports construct validity, inferential reliability, and replicability (Creswell & Clark, 2017). By spanning perceptual, affective, and enacted trust indicators, it allows us to trace not only how learners judge pedagogical avatars, but whether such judgments translate into epistemic compliance—a central question in both trust research and AI-mediated education. To support methodological transparency and reproducibility, we provide a computational notebook with accompanying documentation that implements the end-to-end analytic workflow, archived on Zenodo (Anthis & Kyriakidou-Zacharoudiou, 2026).

5. Results

To evaluate how avatar identity attributes shaped learners' perceptions and behavioral trust across instructional contexts, we present integrated findings from both experiments, highlighting replication fidelity, divergence patterns, and domain sensitivity. Study 1 offered high internal control via a within-subjects design ($N = 102$), while Study 2 extended ecological generalizability through a between-subjects online format ($N = 294$). Analyses focused on four primary outcomes—credibility, warmth, competence, and willingness to act—and incorporated both self-report measures and a behavioral indicator of epistemic uptake. Avatar race (White, Black, Asian), gender (male, female), and age (young, older) were fully crossed, and all outcomes were examined across four instructional domains: STEM, interpersonal, procedural, and reflective.

Results are structured to address the three research questions in turn. We first report the main effects of avatar identity attributes on trust-related evaluations (RQ1). We then examine whether these effects are modulated by learner–avatar identity alignment (RQ2). Finally, we assess domain sensitivity and intersectional penalties associated with compound identity cues (RQ3), followed by an analysis of how these evaluations are reflected in learners' task responses.

5.1. Main effects of avatar identity (RQ1)

Across both studies, avatar race exerted a robust and consistent influence on all perceptual outcomes. In Study 1, repeated-measures ANOVAs revealed significant main effects of race on credibility, $F(2, 202) = 19.43$, $p < .001$, $\eta^2 = .16$, with White avatars rated highest ($M = 4.31$), followed by Asian ($M = 4.02$) and Black avatars ($M = 3.72$). This pattern replicated in Study 2, though with attenuated effect sizes: $F(2, 291) = 10.31$, $p < .001$, $\eta^2 = .066$ (White $M = 4.12$; Asian $M = 3.88$; Black $M = 3.66$). Warmth and competence ratings mirrored this hierarchy, suggesting that race-based perceptual biases generalized across multiple epistemic and affective dimensions (see Fig. 3).

Gender effects were more variable. In Study 1, male avatars were

rated significantly more credible ($M = 4.28$) and competent ($M = 4.27$) than female avatars ($M_{cred} = 3.81$; $M_{comp} = 3.89$), both $ps < .001$. Willingness to act followed a similar pattern. However, in Study 2, gender influenced competence ($F(1, 292) = 4.67$, $p = .032$) but not credibility ($p = .18$), and no significant differences emerged for warmth or willingness. This partial replication suggests that gendered trust biases may be more contingent on controlled exposure and/or task framing. Age yielded more context-sensitive effects. In Study 1, younger avatars received higher credibility ratings overall ($M = 4.12$ vs. 3.79), $F(1, 101) = 9.78$, $p = .002$, $\eta^2 = .07$. However, warmth ratings favored older avatars in reflective and interpersonal domains ($F_{interaction} = 5.02$, $p = .027$), indicating domain–age interplay. In contrast, Study 2 revealed no significant age effects across any outcome ($ps > .10$), implying that age-related biases may require comparative exposure or sustained interaction to manifest. Overall, age-related differences were found contingent on gender (see Fig. 4), indicating context-sensitive variation within avatar attributes; more fully compounded identity effects involving race are examined separately in the subsequent sections.

5.2. Identity alignment effects and social heuristics (RQ2)

Participant–avatar identity alignment influenced perceptual and behavioral responses, though effects varied by identity dimension and study design. In Study 1, within-subject comparisons showed that participants rated avatars who shared their race significantly higher on credibility and warmth ($ps < .001$), with the largest effect observed among Asian and Black respondents. Gender alignment yielded weaker effects ($ps < .05$), and age congruence predicted warmth but not credibility or competence. In Study 2, racial ingroup alignment significantly predicted warmth ($\beta = .29$, $p = .004$) and willingness to act ($\beta = .23$, $p = .009$), but not credibility ($p = .14$) or competence ($p = .09$). Gender and age alignment were non-significant across all outcomes ($ps > .10$), suggesting that race is the most socially salient and automatically processed cue during one-shot, avatar-mediated interactions. The diminished influence of gender and age match in Study 2 may reflect lower identity salience or reduced comparison opportunities in between-subjects exposure (see Fig. 5).

These findings indicate that identity alignment selectively amplifies trust judgments when socially salient cues are available, particularly race. However, alignment effects alone do not account for how trust evaluations shift across instructional contexts or compound across multiple identity dimensions.

5.3. Domain-specific patterns and modulation of bias (RQ3)

Across both studies, instructional domain significantly modulated the expression of identity biases. STEM and procedural tasks elicited stronger race- and gender-based differentiation, while interpersonal and reflective tasks amplified warmth-based judgments. In Study 1, STEM items produced the largest competence differential: Asian male avatars

	Study 1: Controlled (Within-Subjects)			Study 2: Naturalistic (Between-Subjects)		
Credibility	4.31	4.02	3.72	4.12	3.88	3.66
Warmth	4.32	3.92	3.49	3.86	3.46	3.51
Competence	4.40	3.97	3.67	4.27	3.78	3.47
Willingness to Act	4.12	3.81	3.23	3.77	3.80	2.88
	White	Asian	Black	White	Asian	Black

Fig. 3. Perceptual trust ratings by avatar race across studies. Mean ratings on credibility, warmth, competence, and willingness to act. Left: Study 1 (within-subjects). Right: Study 2 (between-subjects). Study 1 reveals sharper race-based differentiation; Study 2 shows slightly reduced separation.

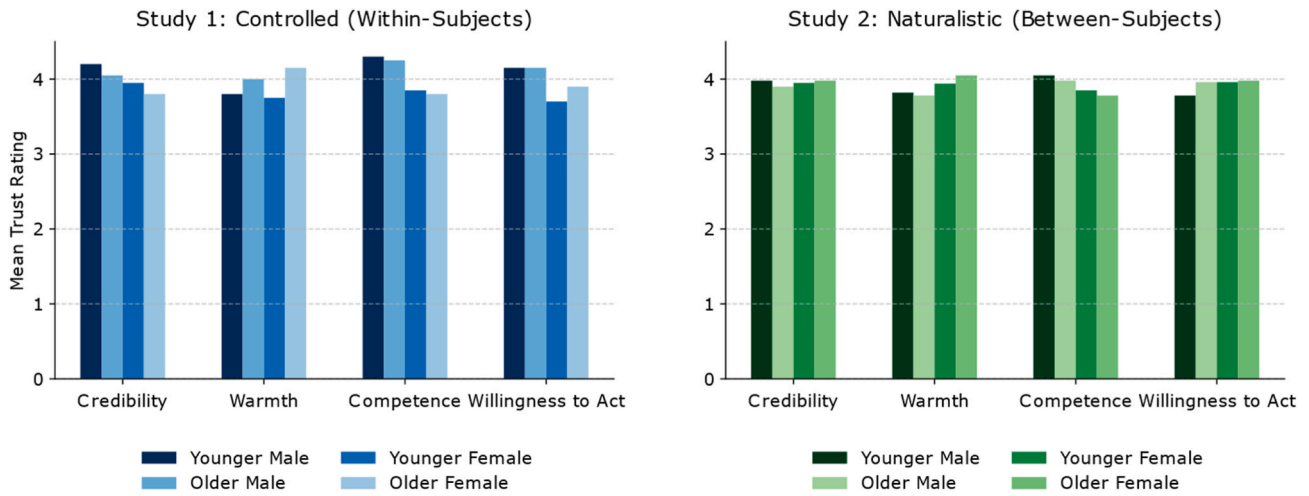


Fig. 4. Trust ratings by avatar Gender x Age. Mean ratings on credibility, warmth, competence, and willingness to act. Left: Study 1 (within-subjects). Right: Study 2 (between-subjects). Gender effects were pronounced in controlled settings, but largely absent under ecologically varied conditions.

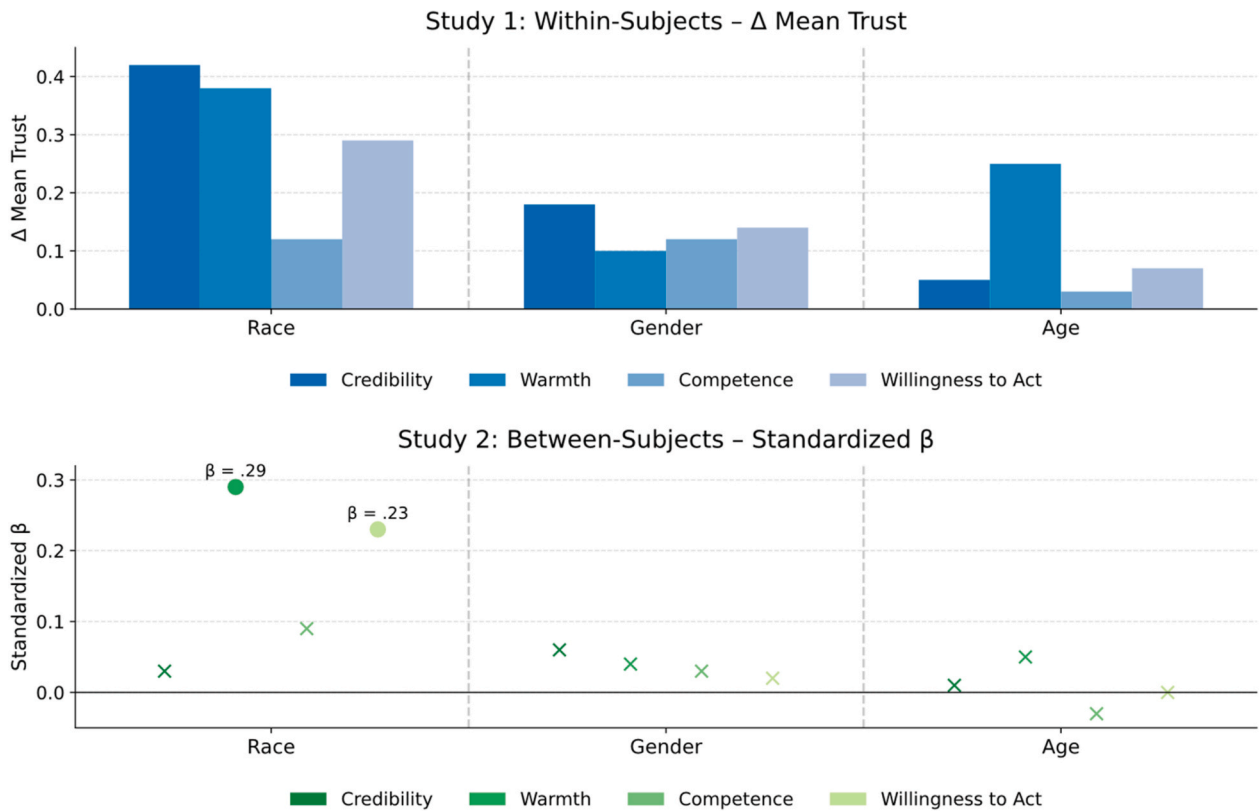


Fig. 5. Effects of avatar-participant identity alignment on trust. Δ Means (top) and standardized β s (bottom) reflect trust differences by alignment on race, gender, or age. Only significant β s ($\geq .20$, $p < .05$) are labeled. Race alignment consistently showed strongest effects, notably on warmth and willingness to act. Colors denote trust constructs.

were rated highest ($M = 4.41$), with older Black female avatars rated lowest ($M = 3.23$), $F_{\text{domain} \times \text{race} \times \text{gender}} = 4.17$, $p = .005$. Interpersonal and reflective domains, by contrast, attenuated race effects but amplified age-warmth linkages. Study 2 echoed these trends with more variability. A race $\times$ domain interaction was significant for credibility, $F(6, 582) = 3.12$, $p = .006$, with race effects most pronounced in STEM and procedural items. However, domain did not moderate gender effects ($p = .21$), and age interactions did not reach significance, suggesting that under naturalistic conditions, race remains the most consistently salient identity marker. Aggregating across domains masked important

shifts: while older avatars were disadvantaged in STEM and procedural contexts, they were perceived as warmer and more trustworthy in reflective settings (e.g., moral reasoning or self-assessment tasks), particularly among older learners (see Fig. 6).

These domain-sensitive shifts indicate that the salience of identity cues is contingent on instructional context; however, they do not capture how multiple identity attributes operate jointly within a single avatar by means of compounded penalties, which is examined next.

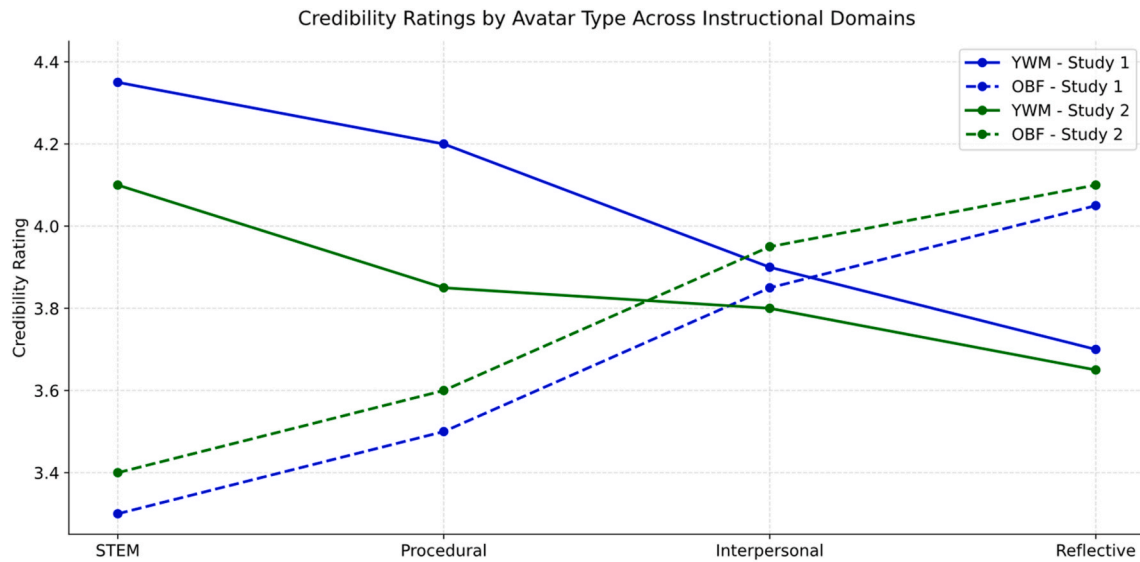


Fig. 6. Credibility rating shifts across domains for two contrasting avatar profiles: Young White Male (YWM) vs. Older Black Female (OBF). Study 1 (solid/dashed blue) shows higher control-induced divergence; Study 2 (solid/dashed green) reflects attenuated but crossover patterns.

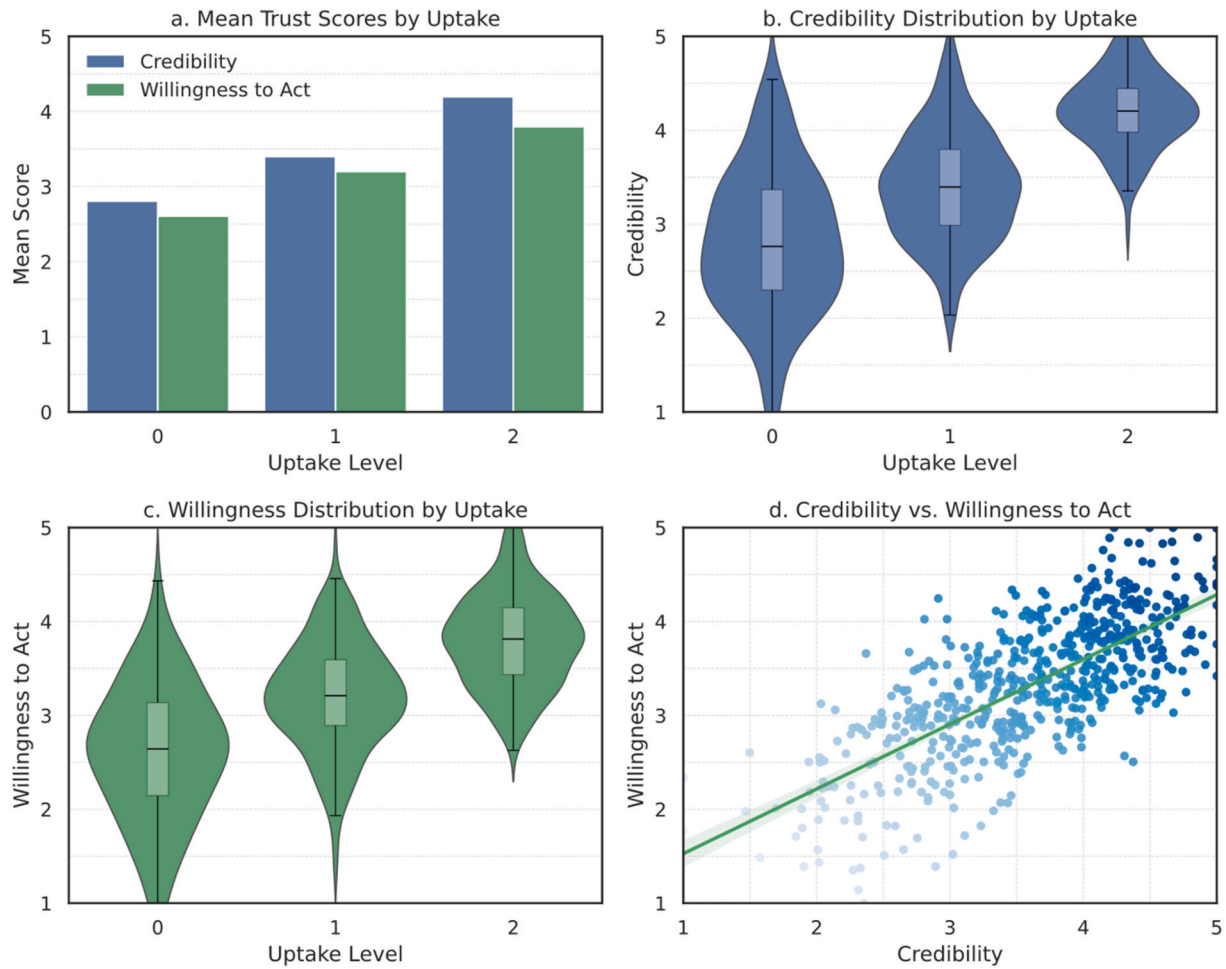


Fig. 7. Relationships between behavioral uptake and trust across both studies. (a) Mean trust ratings by uptake score reveal graded increases in credibility and willingness to act. (b–c) Violin plots illustrate distributional overlap and compression at high uptake levels, with broader dispersion at mid- and low-trust scores. (d) Positive Association between credibility and willingness highlights the link between epistemic and behavioral trust.

5.4. Intersectionality and compounded penalties (RQ3)

Beyond domain effects, both studies revealed additive suppression effects for multiply marginalized avatars; particularly older Black women, who received the lowest aggregate ratings across all constructs. In Study 1, a three-way interaction between race, gender, and age significantly predicted credibility, $F(2, 202) = 4.62$, $p = .011$, $\eta^2 = .05$. Mean ratings for older Black female avatars were 3.48 (credibility), 3.21 (willingness), and 3.37 (competence), consistently below the grand means for each domain. These intersectional penalties persisted in Study 2, though statistical significance was marginal in some cases (race $\times$ gender $\times$ domain interaction for credibility: $p = .062$). Despite greater variance in participant interpretation, the replication of this compounded disadvantage supports intersectionality-informed models of agent perception.

5.5. Behavioral trust and epistemic uptake (Cross-RQ)

A core aim of both studies was to assess whether perceptual judgments translated into behavioral trust. In Study 1, uptake scores (scale: 0–2) strongly correlated with credibility ($r = .62$, $p < .001$) and willingness to act ($r = .54$, $p < .001$). Full uptake (score = 2) occurred in 57.8% of racial ingroup trials compared to 34.2% in outgroup trials, $t(101) = 6.14$, $p < .001$, $d = .71$. Intersectional avatars showed the lowest uptake rates (23.5%), reinforcing the link between social perception and instructional transference. Study 2 showed lower overall uptake ($M = 1.03$ vs. 1.32 in Study 1), likely due to reduced immersion and a single avatar interaction. Still, uptake remained correlated with credibility ($r = .48$, $p < .001$), and ingroup-aligned avatars, particularly White males in STEM, achieved the highest transference rates (43.7% scored “2”). Black female avatars again recorded the lowest uptake (19.1%). Multinomial logistic regression confirmed race and domain as significant predictors of uptake ($ps < .01$), while age and age match were non-significant ($p > .40$), emphasizing the selective influence of salient heuristics (see Fig. 7).

6. Discussion

These experiments reveal a striking paradox at the heart of AI-mediated learning: the very mechanisms designed to make digital tutors more human, avatars, faces, and voices, also reintroduce the same social hierarchies they were meant to transcend. Across both studies, learners consistently judged pedagogical agents not only by the quality of their explanations, but also by the visible social cues those agents displayed. These results confirm a central insight from decades of social cognition research: that judgments of credibility and expertise are shaped by stereotypes, even in digital environments that appear to level the playing field.

These patterns align with Social Agency Theory and the Computers as Social Actors (CASA) framework, which posit that social cues in multimedia environments activate interpersonal schemas, leading learners to treat digital agents as social partners rather than neutral instructional tools (Mayer, 2014; Reeves & Nass, 1996). In parallel, the cognitive theory of multimedia learning highlights how features such as voice, embodiment, and personalization foster perceptions of social presence and epistemic authority, shaping learners’ trust, motivation, and evaluative judgments (Clark & Mayer, 2023; Mayer, 2024). Viewed together, these perspectives suggest that differential responses to avatar race, gender, and age are not anomalous design artifacts, but predictable social-cognitive processes through which learners treat AI tutors as social actors whose perceived authority and trustworthiness must be inferred.

6.1. Identity cues as gatekeepers of trust and action

Our findings extend a body of work that has documented the

persistence of offline social heuristics in avatar-mediated contexts (Edwards et al., 2019; Nowak & Rauh, 2008; Oh et al., 2016). Consistent with these studies, race was the most influential cue, with White avatars receiving higher ratings of credibility and competence. Gender and age effects were more nuanced but still evident, particularly in STEM-related tasks. However, our results go beyond prior research by demonstrating that these perceptual biases shape not just impressions but proximal learning behavior in task performance. Participants were more likely to incorporate guidance into their work when it came from avatars that fit their preconceived notions of credibility. In other words, identity cues functioned not only as evaluation filters but as gatekeepers of epistemic uptake, influencing which ostensibly identical explanations were allowed to shape learners’ own productions. Importantly, this indicates that identity cues do not only influence evaluation, but also shape whether guidance is acted upon. In this respect, the present study offers a bridge between research on trust as a judgment and research on trust as enacted behavior.

These findings indicate that reliance in AI-mediated learning unfolds as a multi-stage process linking perception, intention, and enacted use. Learners’ stated willingness to act did not always map directly onto their observed epistemic uptake, suggesting that evaluative judgments, behavioral intentions, and actual knowledge use can diverge under conditions of epistemic opacity. This divergence is especially consequential in contexts where learners cannot independently verify instructional quality and must decide whether to integrate guidance on the basis of socially mediated cues. More precisely, the study isolates a behavioral “follow-through gap” that becomes visible only when trust is examined alongside uptake, and thereby complements calibration models that focus on alignment between trust and system reliability.

This behavioral dimension also adds nuance to the theory of intersectionality in digital contexts (Crenshaw, 2013). While previous work has shown additive effects of multiple marginalized identities on credibility ratings, our findings show that these compounded disadvantages also diminish the uptake of instructional guidance, with older Black female avatars facing the steepest barriers. This directly addresses whether compounded identity signals shape both learners’ impressions and their willingness to use AI-provided explanations.

From a sociocultural perspective, these effects can be interpreted as differences in whether learners position an AI tutor as a legitimate more knowledgeable other within their zone of proximal development (Vygotsky, 1978; Wertsch, 1991). Within this framework, learning is mediated through socially recognized sources of authority and guidance, rather than merely through exposure to information (Rogoff, 1990). Avatars perceived as competent and credible were more readily granted epistemic authority, while others were implicitly excluded from learners’ learning processes despite equivalent instructional content. Willingness to act and epistemic uptake thus reflect not only trust judgments, but the extent to which learners permit an AI tutor to function as an instructional partner in their learning processes (Luckin, 2018). This framing helps explain why identical explanations can have unequal instructional force, as uptake depends on who is treated as eligible to “teach.”

6.2. Context matters: when domain expectations meet bias

Domain specificity emerged as a critical moderator of avatar-mediated trust. Whereas prior work has often treated trust in pedagogical avatars as a generalizable phenomenon, our results indicate that bias patterns vary systematically with the epistemic demands of the task. STEM and procedural domains, where technical expertise and correctness are most salient, amplified the privileging of young, White, and Asian male avatars; indeed, the largest competence differentials were observed between young White and older Black female avatars on STEM items. Conversely, reflective and interpersonal tasks attenuated competence-based hierarchies and, in some cases, favored older avatars perceived as warmer or wiser.

These patterns suggest that learners dynamically reweight social cues as a function of domain expectations, drawing more heavily on culturally entrenched markers of epistemic authority when tasks foreground technical expertise, and shifting toward warmth-related heuristics when tasks emphasize interpretation or interpersonal judgment. From a design perspective, this implies that equity risks in avatar-mediated learning are context-sensitive rather than uniform, and that avatar design should therefore be domain-calibrated rather than globally fixed. More concretely, bias mitigation efforts may be most consequential in STEM and procedural contexts, where competence heuristics exert the strongest downstream effects on both trust judgments and epistemic uptake. Together, these findings clarify not only *when* identity cues become more consequential, but *under what instructional conditions* they are most likely to shape behavior.

These domain-sensitive effects are also consistent with long-standing theories of epistemic cognition and situated learning, which emphasize that learners' judgments about knowledge and authority are shaped by contextual cues about what counts as "knowing" in a given domain. Research on personal epistemology shows that when tasks emphasize correctness, formal procedures, or canonical solutions, learners are more likely to adopt absolutist or authority-oriented stances, privileging sources perceived as expert or credentialed; by contrast, interpretive or reflective tasks invite more evaluativist views, in which interpersonal understanding and experiential judgment carry greater weight. From this perspective, the observed amplification of competence-based bias in STEM and procedural contexts reflects not merely social stereotyping, but a contextually rational—if socially distorted—alignment between domain expectations and learners' epistemic strategies. Avatar identity cues thus function as situated signals of epistemic authority, whose influence intensifies precisely when instructional contexts encourage deference to expertise. This framing situates avatar-mediated trust within established educational theory, clarifying how identity cues become consequential not because learners abandon epistemic judgment, but because they apply it under uncertainty, using culturally available heuristics.

6.3. Ethical and practical implications

The implications of these results go beyond human-computer interaction. They highlight a fundamental ethical challenge in AI-mediated education: by anchoring trust in surface cues, learners may privilege explanations not because they are correct but because they come from avatars that conform to societal stereotypes of expertise (Moss-Racusin et al., 2012). Such patterns risk entrenching bias and creating subtle but systematic inequities in digital classrooms. Rather than acting as neutral conduits for knowledge, avatars become gatekeepers of epistemic access.

More broadly, these dynamics raise concerns about epistemic injustice in AI-mediated learning environments. When identical instructional explanations are granted different levels of credibility and uptake based solely on the perceived identity of an AI tutor, certain epistemic agents are systematically discounted as knowers, constituting a form of testimonial injustice (Fricker, 2007). At scale, such processes risk embedding longstanding patterns of epistemic injustice into educational technologies. The key risk is not only unequal perception, but unequal access to instructional support, because trust allocation becomes a prerequisite for learning support to count.

These findings therefore raise a critical design question: how can we build AI-mediated learning systems that elicit trust appropriately without reinforcing social hierarchies? Our evidence shows that relying on anthropomorphism alone is insufficient and potentially counterproductive, as it amplifies rather than reduces biases.

6.4. From perception to behavior: recommendations for EdTech

While prior work has proposed interventions such as stylization,

diversity, and transparency to mitigate biased perceptions in avatar-mediated systems (Miller, 2019; Nowak & Rauh, 2008; Ratan & Sah, 2015; Shin, 2021), the present findings demonstrate that identity-based bias operates at the level of action as well as evaluation. Because these biases shape learners' decisions to use or ignore guidance, interventions should be assessed primarily by their effects on guidance uptake (enacted use), not solely by changes in attitudes or even self-reported trust.

To clarify how these implications translate into practice, we distinguish between *technology-facing* considerations, which concern how AI-driven systems are designed and instrumented, and *education-facing* considerations, which concern how AI-driven systems are deployed, i.e., integrated into curriculum, pedagogy, assessment, and institutional policy.

a. Technology-facing considerations

Several strategies emerge for the design of AI-mediated learning systems. Prior research steadily finds that stylized or less anthropomorphic avatars can reduce the automatic activation of stereotypes by limiting the salience of demographic cues (Banakou et al., 2020; Mull et al., 2015; Nowak & Rauh, 2008). Such designs, however, should be carefully phased and paired with consistent behavioral reliability to avoid diminishing engagement. Rotating diverse avatars across tasks can disrupt the association of competence with a single demographic profile, particularly when rotations are calibrated by domain to address context-specific stereotypes (Banakou et al., 2020; Dasgupta & Greenwald, 2001; Ratan & Sah, 2015). Transparency and provenance cues—including explanations of content generation and visual signals of uncertainty—can further redirect attention from surface appearance to epistemic evidence (Miller, 2019; Shin, 2021). In parallel, trust calibration tools such as interactive explanations, provenance indicators, and reflection prompts can support learners in evaluating guidance based on epistemic quality rather than socially patterned cues (Lee & See, 2004; Miller, 2019).

Bias mitigation must also be addressed earlier in the data-to-design pipeline. Fairness-aware generation methods, participatory content development, and sociotechnical alignment protocols can promote equitable representation from the earliest stages of AI tutor development, shifting accountability from post hoc filtering to design-stage decisions. At the level of learning environments, these choices shape the social-epistemic climate of digital classrooms by distributing epistemic authority unevenly across embodiments. When certain avatars are consistently perceived as more credible or competent, learning environments may develop patterned concentrations of epistemic authority (Rogoff, 1990; Veletsianos, 2010; Vygotsky, 1978), potentially undermining inclusivity or fairness, even if instructional content is held constant.

These dynamics have direct implications for educational data mining and learning analytics. Behavioral indicators such as help-seeking, feedback incorporation, epistemic uptake, and revision trajectories—often treated as neutral proxies for engagement, mastery, or persistence—may partly reflect learners' epistemic positioning toward particular embodiments rather than learning progress alone (Charland et al., 2024; Gašević et al., 2015). In adaptive learning systems, recommender pipelines, and LMS-driven personalization, socially patterned reliance can therefore be folded into predictive and prescriptive models, reinforcing biased learning trajectories unless analytic systems explicitly account for identity-mediated trust and uptake processes. Institutions should monitor not only aggregate performance but also the distribution of trust, reliance, and uptake across avatar configurations and learner groups, auditing for emergent epistemic imbalances and validating adaptive systems across socially diverse learners.

b. Education-facing considerations

Crucially, these design choices are not external to pedagogy. AI tutors function as curricular voices rather than neutral delivery mechanisms. When identical explanations are differentially trusted and enacted, curriculum is shaped not only by content sequencing but by epistemic voice. Instructional strategies that incorporate AI tutors therefore entail implicit decisions about instructional authority (Mayer, 2014; Reeves & Nass, 1996). Because learners respond to AI tutors as socially embedded instructional partners, instructional effectiveness depends not only on explanation quality but on whether learners accept the tutor's guidance as legitimate. Supporting epistemic vigilance, reflective engagement, and critical evaluation of AI guidance should thus become explicit objectives of AI-mediated learning.

At an institutional level, decisions about default avatar configurations, representation policies, and deployment contexts constitute pedagogical and ethical choices rather than purely technical ones. Educational technology providers and adopting institutions should establish explicit design guidelines, evaluation criteria, and accountability mechanisms to ensure that identity-related design features do not introduce systematic disparities in instructional effectiveness. Translating these principles into practice requires embedding fairness-oriented safeguards within institutional infrastructures, including policy guidelines, compliance mechanisms, and socio-technical governance models for AI tutor deployment. Alignment with international frameworks for AI literacy and ethics (e.g., UNESCO, 2022; OECD European Commission AI Literacy Framework, 2025) can support policy legitimacy and transferability across contexts.

Finally, these findings foreground AI deployment in education as a form of *epistemic governance* (Fricker, 2007). Procurement decisions, default configurations, evaluation criteria, and vendor accountability mechanisms actively shape whose explanations are amplified and acted upon at scale. When such mechanisms remain underdeveloped, epistemic inequities are not merely tolerated but systematically produced through routine institutional practice. Responsible governance must therefore extend beyond technical accuracy and explainability to include standards for identity-sensitive design, reliance monitoring, and transparency practices that protect equitable access to instructional support. Without such guardrails, AI-mediated learning risks reproducing socially patterned inequalities under the guise of personalization and innovation.

6.5. Balancing trade-offs and setting future directions

Translating these systemic commitments into design practice raises practical tensions. These interventions are not without trade-offs. Less anthropomorphic avatars may reduce bias but also diminish engagement and social presence. Rotating diverse avatars may weaken stereotyping but risk inconsistency in the tutor–learner relationship. Transparency and provenance cues can enhance epistemic focus, yet excessive information may reduce usability. A key next step is to evaluate these interventions not only for attitudinal shifts but for sustained changes in guidance uptake and learning outcomes. This requires testing in authentic classroom settings, over extended periods, and across diverse cultural contexts to determine which strategies reduce bias while preserving engagement. Three methodological limitations should be noted here. First, epistemic uptake, operationalized via semantic alignment, indexes the *transfer of information into output* but does not necessarily reflect its *internalization into conceptual understanding*. Such alignment may capture compliance or surface-level paraphrasing as much as genuine knowledge appropriation. Second, the overall validity of the construct could be further strengthened by calibrating semantic similarity thresholds against alternative metrics such as knowledge convergence or dialogic quality indicators. Third, although the findings are consistent with a process in which avatar identity shapes trust, which in turn informs willingness and uptake, this pathway remains inferential. Further specifying these dynamics requires formal modeling (e.g., mediation or structural approaches) to determine whether trust

dimensions act as causal intermediaries or co-occurring correlates of behavior, advancing from descriptive association to mechanistic explanation.

Beyond these constraints, the findings point to several directions for further inquiry. One priority is to examine how learners' epistemic positioning toward AI tutors evolves across repeated interactions, and how early impressions translate—or fail to translate—into sustained uptake. Greater precision is also needed to disentangle dimensions of trust; decomposing credibility, warmth, and competence into finer components may support more targeted tutor personalization and equity-focused design, particularly where learners experience cumulative disadvantage. Perspectives of marginalized learners warrant explicit attention. A related line of inquiry concerns how epistemic injustice and epistemic oppression (Dotson, 2012; Fricker, 2007) manifest in AI-mediated learning. Examining stereotype threat, testimonial silencing, and differential interpretability will be critical for understanding variation in engagement. Cross-cultural and multi-site comparative studies, including variation in identity cues such as accent or linguistic style, alongside longitudinal analyses of repeated exposure to diverse avatars, are needed to assess generalizability beyond controlled settings. Together, these efforts will clarify how, when, and for whom AI learning environments can foster equitable and trustworthy educational interactions.

In sum, our findings suggest that mitigating bias in AI-mediated education requires moving beyond cosmetic changes and focusing on how learners act on guidance. By embedding additional mechanisms that deliberately shape how trust is formed and enacted, future systems can ensure that digital classrooms amplify content rather than stereotypes (Holmes et al., 2022).

7. Conclusions

All in all, this study foregrounds a critical tension in AI-mediated education: interface features intended to foster pedagogical rapport may simultaneously reproduce socially patterned biases. Findings make clear that bias in AI-mediated education is not an algorithmic defect alone; it is a social phenomenon that emerges wherever human cognition meets digital embodiment. Race proved to be the most salient cue, with compounded disadvantages for older Black female avatars, and these judgments translated into real behavioral differences in whether guidance was accepted or ignored. Without intervention, the digital classroom will silently reproduce the same asymmetries of trust and influence that characterize its offline counterpart.

The implications extend beyond the technical. Designing future AI tutors will require attention not only to accuracy and adaptivity, but to equity; decisions about how avatars look, speak, and behave are ethical decisions about whose voices are amplified and whose are diminished. The promise of AI in education will not be fulfilled by more powerful models alone, but by systems deliberately built to confront bias, so that the face of AI becomes a bridge to inclusion rather than a mirror of exclusion.

At the same time, these findings must be interpreted with caution. The present work focused on a limited set of visible identity cues, and while the present sample included racially and demographically diverse participants, it was drawn primarily from a WEIRD (Western, Educated, Industrialized, Rich, and Democratic) (Henrich et al., 2010) context, and consisted largely of English-speaking university populations, which constrains the cultural and ecological generalizability of the results. As constructs such as credibility, warmth, and epistemic uptake are known to be shaped by cultural norms and epistemological traditions (Hofstede, 2016; Kirmayer et al., 2020), the patterns observed here may not transfer directly to learners from different cultural backgrounds. The experiments also relied on relatively brief, controlled interactions with avatars, and future work will need to examine whether the same patterns hold in more complex, longer-term classroom and collaborative settings. Trust and epistemic uptake were modeled as static outcomes,

rather than as dynamic processes that may evolve over repeated interactions. In addition, the study did not incorporate multimodal features (e.g., voice, gesture) that increasingly accompany avatar-based AI tutors, nor did it capture how learners' prior experiences or critical digital literacy might shape their resistance to bias. Taken together, however, these constraints point directly to the next phase of inquiry. They provide a foundation for extending this work across diverse cultural contexts, multimodal environments, and longitudinal settings, where trust can be examined as an evolving, context-sensitive process. The findings offer a starting point for investigating how AI-mediated learning environments can support more calibrated, equitable, and reflective forms of epistemic engagement.

CRedit authorship contribution statement

Zach Anthis: Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization, Writing – review & editing. **Avgousta Kyriakidou-Zacharoudiou:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization, Formal analysis.

Informed consent statement

Informed consent was obtained from all subjects prior to their participation in the study.

Institutional review board statement

The study was conducted in accordance with GDPR and approved by the Institutional Review Board (protocol code 775/2025).

Data availability statement

The datasets generated and analyzed during the current study are not publicly available due to ethical and privacy considerations related to human-subject research and the sensitive nature of the demographic and identity-related variables examined.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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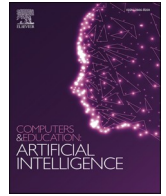
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Update

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Corrigendum to “Face value: How avatar identity shapes epistemic trust in AI-mediated learning” [Computers and Education: Artificial Intelligence 10 (2026) 100610]

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The authors regret that they have noticed the following two issues in the final publication of their article, “**Face Value: How Avatar Identity Shapes Epistemic Trust in AI-Mediated Learning**” (CAEAI_100610) and would be very grateful if it would be possible to please correct them.

1. First, we kindly ask that the following sentence:

“To support methodological transparency and reproducibility, we provide a computational notebook with accompanying documentation that implements the end-to-end analytic workflow, archived on Zenodo and to be made publicly available upon acceptance (DOI: xx.xxxx/zenodo.xxxxxx).”

be replaced with:

“To support methodological transparency and reproducibility, we provide a computational notebook with accompanying documentation that implements the end-to-end analytic workflow, archived on Zenodo (Anthis & Kyriakidou-Zacharoudiou, 2026).”

The corresponding reference is:

Anthis, Z., & Kyriakidou-Zacharoudiou, A. (2026). Replication Package for “Face Value: How Avatar Identity Shapes Epistemic Trust in AI-Mediated Learning”. Zenodo. <https://doi.org/10.5281/zenodo.18445641>.

We apologize for this oversight. This correction is requested because the current text still contains a placeholder DOI and acceptance-stage wording, whereas the Zenodo record is now publicly available with a permanent DOI.

2. Second, we noticed **two** uncited references that were not initially identified by the platform.

- **The first is Schemmer et al. (2022)** in Section 2.5, “Willingness to act and epistemic uptake: from perception to behavior,” in the following sentence:

“However, a robust finding across automation and decision-support research is that intention is not equivalent to reliance. Users may report high trust and strong willingness to follow system outputs while simultaneously engaging in inconsistent, minimal, or strategically selective use in practice (Dzindolet et al., 2003; Lee & See, 2004; Schemmer et al., 2022).”

The citation should be corrected to Schemmer et al. (2023), and the corresponding reference should be:

Schemmer, M., Kuehl, N., Benz, C., Bartos, A., & Satzger, G. (2023, March). Appropriate reliance on AI advice: Conceptualization and the effect of explanations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (pp. 410–422).

- **The second is Hendriks et al. (2015)** in Section 2.2, “Competence as perceived instructional authority,” in the following paragraph:

“Within the METI framework, this dimension corresponds to trustee expertise, understood as the perceived capacity of a system to reliably execute domain-relevant tasks, even when supported by heuristic or pattern-based processes (Hendriks et al., 2015).”

While the corresponding reference is correctly included in the reference list at the end of the paper, this particular citation is not hyperlinked.

Hendriks et al. (2015) is also not hyperlinked in Section 2.3 “Credibility as Epistemic Reliability” in the following paragraph:

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“Within the present framework, credibility—aligned with METI’s integrity—thus represents the learner’s evaluative verdict on whether an instructional agent constitutes a legitimate epistemic source, grounded not in presumed intent nor mere computational performance, but in perceived epistemic reliability and responsible

knowledge deployment (Hendriks et al., 2015; Durán & Pozzi, 2025; Ferrario, 2024).”

The authors sincerely apologize for any inconvenience this may cause.